LESSON GUIDE
Technology Plus
Unit 7: Security
6.1.7 – AAA Accounting

Lesson Overview:
In this lesson, students will explore accounting regarding cybersecurity, and why it is essential for accountability, detecting suspicious activity, and non-repudiation.

Students will:
- Explain what accounting is regarding cybersecurity, and why it is important.
- Describe how logs, location tracking, and web browser history are used to record user actions.
- Analyze digital logs, location data, and browser history to identify suspicious or unauthorized activity.

Guiding Question: How do computers keep track of what users do to ensure accountability and security?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 45-70 minutes

Standards:
CompTIA Tech+ FC0-U71 Objective

- 6.1 - Explain fundamental security concepts and frameworks.
  - Authentication, authorization, accounting, and non-repudiation concepts
    - Authentication
      - Single factor
      - Multifactor
      - Single sign-on
    - Authorization
      - Permissions
    - **Accounting**
      - **Logs**
      - **Location tracking**
      - **Web browser history**
  - Administrator vs. user accounts
      - Least privilege model

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).

Lesson Background

Background Information:

In cybersecurity, the AAA triad – the third A stands for accounting which ensures that everything a user does is recorded, tracked, and reviewed.

Accounting is sometimes called auditing because it creates a digital trail of activity. This trail is critical for maintaining security, spotting suspicious behavior, and holding people accountable. Without accounting, unauthorized actions might go unnoticed and users could deny their involvement – making it nearly impossible to investigate incidents.

Some common examples of accounting include: Logs: Records of logins, file access, and changes to a system. Location tracking: Identifying where a user or device was when they accessed a system. Web browser history: Showing what sites or resources a user visited, which can help identify risky or suspicious behavior.

Accounting also supports the related security concept of non-repudiation, meaning users cannot deny the actions they've taken because there's evidence in the logs. For example, if a file is deleted at a certain time, the logs can prove who deleted it, where they were, and what else they were doing online. By learning about accounting, students gain insight into how organizations detect breaches, investigate incidents, and build trust in digital systems. In this lesson, students will step into the role of a cybersecurity investigator, using logs, location data, and browser history to piece together what happened during a possible security incident.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.1.7 – AAA Accounting • Activity 6.1.7 – Tracking Trouble
Materials per Student	• Student Notes 6.1.7 – AAA Accounting • Student Handout 6.1.7 – AAA Accounting • Assessment 6.1.7 – AAA Accounting (optional)

Lesson Background

Cyber Terms:

Term	Definition
AAA Triad	a cybersecurity framework that secures access to systems and data by making sure users are who they say they are (Authentication), can only do what they're allowed to do (Authorization), and that their actions are recorded (Accounting)
accounting	often referred to as auditing, involves tracking and recording user activity on a system to ensure compliance and detect inconsistencies
logs	digital records that computer systems automatically create to document user actions, system events, and transactions
location tracking	monitors and records the physical or network location of users or devices, typically based on IP addresses, GPS data, Wi-Fi, and other network information
geofencing	a virtual boundary around a real-world location
web browser history	keeps a record of the websites and pages a user visits

6.1.7 – AAA Accounting

Guiding Question: How do computers keep track of what users do to ensure accountability and security?

Lesson Launch (10-15 minutes)

- **Ask:** If an unauthorized user successfully logged into your social media account, how could you prove it wasn't you?
- **Say:** Phones, computers, and networks keep records of activity through logs, location data, and web browser history.
- **Mini Activity:** Have students check the browser history on their device.
  - **Ask**: What does this history say about your actions? What story does it tell?
- Remind students of the AAA Triad and tell them that today, they will learn about the third A – **accounting** which is about keeping track of what users do on a system.
  - Accounting in the AAA Triad does not have anything to do with money or math.
- **Ask:**
  - Have you ever gotten an alert saying, 'We noticed a login from a new device.'?
  - Has your streaming service remembered exactly where you stopped in a show?
  - Has a mobile app documented the exact time you completed an action?
- All those things are accounting. It creates a trail of evidence showing who did what, when, and where.

Digital Footprints (15-20 minutes)

- Distribute the Student Handout 6.1.7 – AAA Accounting and have students fill in the answers during the lesson.
- Discuss ***accounting***, its definition, and benefits.
  - Accounting ensures there is a record of actions so people can be help responsible and suspicious activity can be detected.
  - Benefits of Accounting in the AAA Triad:
    - Accountability: Users can be held responsible for their actions.
    - Non-Repudiation: People can't deny their actions later.
    - Security Monitoring: Helps detect suspicious or malicious activity.
    - Troubleshooting & Recovery: Logs help IT staff figure out what went wrong and fix problems.
    - Evidence: Provides proof in case of rule violations or cybercrimes.
  - Accounting is like having security cameras for digital activity making a system safer, fairer, and more reliable.

- Introduce ***logs***, their definition, the types of logs, and examples.
    - Types of Logs:
        - Login/Authentication Logs: Record when users log in, log out, or fail to log in.
            - Shows if someone tried (and failed) to guess your password 10 times in a row.
        - Access Logs: Track what file, apps, or websites were accessed.
            - A log can show that you opened the school's shared drive at 10:32 AM.
        - System/Event Logs: Document changes to a system (software updates, errors, crashes).
            - Shows that a computer shut down suddenly at 2:45 PM due to overheating.
        - Network Logs: Record traffic across a network (what devices connected, what data was sent).
            - Helps track a hacker trying to sneak data out of a company.
    - Logs are like a receipt book for computers, every action leaves a receipt of who did it, what they did, and when it happened.
- Discuss ***location tracking*** and ***geofencing***, their definitions, and how they relate to accounting.
- Discuss ***web browser history***, its definition, and how it assists the AAA Triad.
    - Web browser history ensures:
        - Authentication: Browsers of the store credentials and session information for websites.
        - Authorization: Browsers can control access to resources through permissions, such as allowing or blocking access to location, camera, or microphone.
        - Accounting: Browsers maintain history and logs of visited websites, downloads, and interactions.
        - Privacy: Browsers offer settings to manage cookies, site data, and tracking.

Activity: Tracking Trouble (15-25 minutes)

- Display Activity Slides 6.1.7 – Tracking Trouble, have students record their answers in the handout.
- Group students into pairs or small teams (2–4 students).
- Distribute 6.1.7 – Tracking Trouble Data. This document is the same as slide 3 and includes the login/access logs, location data, and browser history needed for the investigation.

Scenario

- It's a regular school day, and the math teacher, teacher01, uploaded a lesson plan file (Math101.docx) to the school's shared drive during the morning. Later that afternoon, the teacher reported that the file had suddenly disappeared.
- As part of the school's IT team, you suspect someone may have accessed the system without permission. To investigate, you pull logs, location tracking data, and browser history records from the school's network monitoring tools.
- Students should take on the role of the school's IT investigation team. Using the data, they will:
    - Identify unusual patterns in the logs.

- Flag suspicious entries and explain why they are red flags.
- Cross-check evidence across multiple sources (logs, location data, browser history).
- Decide who is responsible for the file deletion and whether it was a legitimate action or a security incident

Activity Debrief

1. What unusual patterns do you notice in the login attempts and file access logs?

    The logs show normal logins from teacher01 at the school, but guest02 logs in from a suspicious IP address and location.

2. Which specific entries raise red flags, why?

    Guest02 successfully logged in, then deleted a teacher's file. Guest accounts aren't supposed to delete teacher files, so this seems like abnormal behavior for a guest account.

3. Which user appears responsible for the suspicious file deletion?

    Guest02 is responsible for deleting file 'Math101.docx'.

4. How can you prove this user is suspicious using more than one source of evidence?

    The access logs show the file deletion, location data shows the login came from New York instead of the school, and browser history shows visits to suspicious websites.

5. Compare the locations of teacher01 and guest02. What does this suggest?

    Teacher01 logged in from the school campus, while guest02 logged in from New York. Since guest02 should not be accessing school files remotely, this suggests unauthorized access.

6. Do the websites visited by each user support normal academic activity or suspicious behavior?

    Teacher01 visited educational websites, showing normal academic behavior. Guest02 visited cheat sites, file-sharing platforms, and hacker forums – clear signs of suspicious behavior.

7. What might have happened if there were no accounting tools in place?

    Without accounting tools, the teacher would notice the file was missing, but there would be no way to trace who accessed it, when it was deleted, or whether it was an accident or an attack.

8. How do logs, location tracking, and browser history work together to create accountability?

    Logs show what happened and when, location tracking shows where it happened, and browser history gives context about user intent.

9. If you were the school's IT administrator, what alert or rule could you set up to catch this suspicious activity faster?

    I would set up alerts for logins from outside the school network and for file deletions by guest or non-teacher accounts.

10. Why would relying on just one of these be less effective?

    Each source gives only part of the picture. Together, they provide stronger evidence. Relying on just one might leave gaps or make it harder to prove who was responsible.

Putting It All Together (5-10 minutes)

Discussion Questions

- Review what students learned today by discussing the questions below as a class.
- Why is accounting important in the AAA Triad, and what benefits does it provide?

  Accounting proves who did what, when, and where. Without it, suspicious actions could be noticed but never proven. Its benefits include holding users accountable, preventing denial of actions (non-repudiation), spotting security threats, helping IT staff troubleshoot problems, and providing evidence when rules or laws are broken.

- How do different types of logs each help create accountability?

  Each log adds a piece of the puzzle: login/authentication logs show who tried to access the system, access logs show what files or apps were touched, system/event logs reveal changes or crashes, and network logs track data flow. Together, they provide a timeline and context for user behavior.

- How do location data and browser history strengthen accounting evidence compared to logs alone?

  Logs record actions, but location data proves where a user was, and browser history shows intent behind their actions. Combined, they make the evidence stronger – not just what happened, but also where and why.